Robust OFDM Channel Modelling for Wi-Fi CSI Sensing in GNU Radio

ECE 310 Wireless Communications | Project Milestone 4 (Final Report) | Team G1 | GSoC / Open-Source Track

Name	Enrollment no.
Hirva Vekariya	AU2440126
Tirth Pathar	AU2440238
Divyaraj Vaghela	AU2440251
Jahnavi Patel	AU2440024
Jinay Shah	AU2440296

Item	Details
Domain	Wi-Fi sensing (CSI from an OFDM receiver)
Track	GSoC / open source, organisation: GNU Radio
Issues addressed	GNU Radio #6891: OFDM loopback drops packets with zero noise (main contribution). GNU Radio #7997: OFDM flowgraph freezes (investigation).
Upstream pull request	gnuradio/gnuradio PR #8271, opened 11 Nov 2026, open and under maintainer review
Issue comment	Measurements posted on #6891, 11 Nov 2026
Software	GNU Radio 3.10.12.0 (conda-forge) and 3.10.9.2 (Ubuntu 24.04 package), Python 3.11 / 3.12
Team repository	g1_gsoc_wifisensing

Abstract

GNU Radio's stock OFDM transmitter and receiver (gr-digital) silently lose about 4 % of back-to-back packets on a perfect, noise-free channel (issue #6891), and OFDM flowgraphs can freeze without raising any error (the symptom reported in issue #7997). Both problems matter for Wi-Fi sensing, because every received packet produces one channel-state-information (CSI) sample: a receiver that silently drops packets produces a CSI time series with hidden gaps.

Across four milestones we built a sequence-numbered measurement harness and obtained six concrete results. (1) The stock chain delivers 95.88 % of packets (103 of 2500 lost) on an ideal channel. (2) Streaming and offline runs lose exactly the same packets in 5 of 5 seeds, so the GNU Radio scheduler is not the cause. (3) With touching bursts the Schmidl & Cox timing trigger varies between 69 and 73 samples after the true burst start, and a one-line model of the Header/Payload Demux predicts every lost packet in 100 of 100 instrumented runs. (4) A transmitter-side gap (rolloff 4, or a single zero sample) restores 100 % delivery in all 20 multipath and SNR conditions we tested; multipath makes the bug worse (88.5 % on a harsh channel). (5) For sensing, 7 to 9 % of CSI samples go missing, which biases a 0.5 Hz motion-frequency estimate by 7 to 8 %. (6) Packets of 1020 bytes or more are never transmitted, and packets of 300 to 1000 bytes freeze the flowgraph with default buffers.

On 11 Nov 2026 we posted our measurements on issue #6891 and opened pull request #8271, which adds the first multi-packet OFDM regression test to GNU Radio. On 12 Nov the maintainer asked whether the example change "abuses the rolloff factor to hide a bug". We agree with him, and have prepared a revised test that documents the bug (@unittest.expectedFailure) without changing the example. This revision has not been pushed yet; it will go into the same PR #8271 once the maintainer confirms. PR #8271 is the only pull request we have submitted. The receiver-side fix is the core of our refined GSoC 2027 proposal.

1. Introduction and Problem Formulation

1.1 Problem

Wi-Fi sensing infers activity, gestures or presence from CSI, the per-subcarrier complex channel estimate that an OFDM receiver computes from each packet's preamble [5-8]. Research tools obtain CSI from closed Wi-Fi chips (Intel 5300, Atheros, Nexmon, ESP32, PicoScenes) [10, 11]. These give real hardware but no control over the physical layer. GNU Radio offers an open and fully controllable OFDM chain, so it could serve as a reproducible CSI source, but two reliability issues stand in the way:

- **Issue #6891, silent packet loss.** The official ofdm_loopback example drops packets even with zero noise. A maintainer called it "a rather serious" bug (Feb 2024), but no measurement, sample-level root cause or regression test existed.
- **Issue #7997, silent freeze.** An OFDM flowgraph stops producing output after a Channel Model block is inserted, without raising an exception.

1.2 Research Question and Objectives

Research question. How reliable is GNU Radio's OFDM chain as a packet and CSI source, why does it fail, and what is the smallest change that GNU Radio maintainers would accept to detect or fix the failure?

Objective	Measurable target	Status
O1. Quantify the loss	Packet delivery ratio (PDR) per seed, per channel and per TX setting	Done (M2, M3)
O2. Separate scheduler from signal processing	Same lost packets in streaming and offline runs	Done: 5/5 seeds identical (M2)
O3. Explain the loss at sample level	Model that predicts which packets are lost	Done: 100/100 runs predicted exactly (M3)
O4. Measure the effect on CSI sensing	Missing CSI fraction and motion-frequency error	Done: 7-9 % missing, 7-8 % bias (M3)
O5. Contribute upstream Issue comment, pull request, maintainer review	**Used** Done: issue comment and one PR (#8271) on 11 Nov; revision prepared after the review, not pushed yet (M4)	

1.3 Wireless-Communication Concepts

ECE 310 concept	Where it appears in this project
OFDM, FFT, cyclic prefix	64 subcarriers, 16-sample CP, 80-sample symbols; a 50-byte packet is 12 symbols = 960 samples
Timing synchronisation	Schmidl & Cox metric and its plateau detector [3]; trigger position measured to the sample

AWGN and SNR	SNR sweeps from 6 to 30 dB, and noise-free runs
Multipath, delay spread, frequency-selective fading	Exponential power-delay profiles with rms delay 0, 35, 100 and 200 ns at 20 MHz [12]
Carrier frequency offset (CFO)	0 to 3.5 subcarrier spacings

Modulation, detection, CRC	BPSK header and payload, CRC32
Channel estimation (CSI)	GNU Radio's estimate from ofdm_chanest_vcvc compared with the true H(f)
Network performance	PDR, lost packets, flowgraph stability

1.4 Scope and Assumptions

- Simulation only; no SDR hardware.
- Stock GNU Radio blocks; no new sensing algorithm.
- The contribution follows GNU Radio's contribution process: CONTRIBUTING guidelines, DCO sign-off on every commit and maintainer review.

2. Related Work

Work	**What it does**	**Gap relevant to us**
Ma, Zhou, Wang 2019 [5] (survey)	CSI-based sensing methods and applications	Assumes reliable CSI; receiver reliability not discussed
Du et al. 2025, IEEE 802.11bf [6]	Standard procedures and KPIs for WLAN sensing	Requires timestamped measurements; implementation-agnostic
Radwan et al. 2026 [7]	Self-supervised learning for Wi-Fi sensing	Pre-processing assumes regularly sampled CSI
SenseFi, Yang et al. 2023 [8]	Benchmark library and datasets	Uniformly sampled CSI assumed
Schmidl & Cox 1997 [3]	Preamble-based timing and frequency synchronisation for OFDM	Analyses one burst at a time; back-to-back bursts not studied
GNU Radio gr-digital OFDM [4]	Generic OFDM TX/RX used in examples and teaching	Our target; its QA tests send only one packet
gr-ieee802-11, Bloessl et al. [9]	802.11a/g/p PHY in GNU Radio	Own synchronisation; no packet-level reliability tests of the kind we built
CSI tools (Intel 5300 [10], Nexmon [11], ESP32, PicoScenes)	CSI from commercial chips	Real hardware, but closed PHY and uncontrollable channel
Issue #6891 discussion	Qualitative explanation (plateau + demux) and a TX padding/rolloff workaround	No measurement separating scheduler from signal processing; no test

3. Methodology

3.1 Measurement Harness and Instrumentation

- **Sequence-numbered packets.** Bytes 0 and 1 of every payload carry the packet index and the remaining bytes are random. Every lost packet is identified exactly (repeated payloads hide losses, as the #6891 thread showed).
- **Two execution modes.** Streaming (TX, channel and RX in one flowgraph) and offline (TX runs to completion, RX runs later on the stored signal). Identical losses in both modes rule out the scheduler.
- **Watchdog.** Every trial runs in a separate process with a timeout, so a frozen flowgraph is recorded as HANG instead of stopping the experiment.
- **Fixed seeds** (1 to 5) for payloads and noise, plus three unscored flush packets at the end. ● **Trigger logging (M3).** The receiver's debug output gives the Schmidl & Cox trigger stream and the per-frame channel estimate; both are compared with the known burst starts and the true channel.
- **Header/Payload Demux replay model (M3).** A trigger is used only if it arrives at least (frame length - 1) samples after the last used trigger; an earlier trigger falls inside the frame that is still being read and is ignored. The logged triggers are replayed through this rule and the rejected packets are compared with the lost ones.
- **Channel model (M3).** Static or per-burst FIR taps (multipath), CFO and AWGN at a defined SNR; a time-varying reflector at 0.5 Hz for the sensing experiment.

3.2 Metrics (Defined Before Running)

Metric	Definition
PDR	Packets with valid CRC and correct sequence number / packets sent
Trigger offset / spacing	Trigger position relative to the true burst start / to the previous trigger, in samples
Prediction accuracy	Lost packets equal the packets rejected by the replay model (same sequence numbers)
Flowgraph stability	OK or HANG within the timeout
CSI NMSE	Sum of $(s\lvert\hat{H}\rvert - \lvert H\rvert)^2$ / sum of $\lvert H\rvert^2$ over the 52 occupied subcarriers, with one global scale s
Motion-frequency error	$\lvert\hat{f} - f_{true}\rvert / f_{true}$ for a 0.5 Hz reflector
SNR	Mean TX sample power / noise power

3.3 Experiments

ID	Milestone	Question answered Size
E1	M2	How large is the loss; scheduler or signal processing; which workaround works 60 trials × 500 packets
E2	M2	Effect of packet length 20 × 500
E3	M2	Behaviour with AWGN, 6 to 30 dB 130 × 300
E4, E5	M2	When does the chain freeze (packet size, leaked length tags) 21 trials
E8, E8b	M3	Sample-level mechanism; why some packet lengths lose 40 × 500, 60 × 300

		nothing
E6	M3	Multipath × SNR 200 × 300
E7 E9	M3 M3	Carrier frequency offset 160 × 300 Effect on CSI and on a sensing estimate 50 × 500
E10	M3	Wi-Fi-MTU-size packets 18 × 20
T1-T3	M2-M4	Upstream test passes with the workaround, fails on the See Section 6.6 bug, and is deterministic

4. Implementation: The Upstream Contribution

4.1 Timeline of Upstream Actions

Date	Action Link
11 Nov 2026	Posted our measurements and root-cause analysis on issue #6891 Issue comment on #6891
11 Nov 2026	Opened pull request #8271 with two signed-off commits; DCO check passed PR #8271
12 Nov 2026	Maintainer marcusmueller reviewed the PR: "are you proposing to abuse the rolloff factor to hide a bug?"; label "Needs Action" PR #8271
14 Nov 2026	Prepared our response to the review: we agree, explain the receiver-side root cause, and propose a revised test for the same PR (not pushed yet) PR #8271

4.2 PR #8271 as Submitted (11 Nov 2026)

Part of GNU Radio	Change
OFDM TX/RX QA test (gr-digital)	Added a helper _txrx_multipacket(n_packets, n_bytes, rolloff) and the test test_005_tx_multipacket_rolloff: 100 back-to-back 50-byte packets with random payloads and rolloff=4; every byte must match
ofdm_loopback example flowgraph	rolloff changed from 0 to 4

Commits: 5c37fab "digital: ofdm: add back-to-back multi-packet loopback test" and 9467c3e "digital: examples: use rolloff=4 in ofdm_loopback", both signed off.

4.3 Maintainer Review and Our Proposed Revision

The maintainer's question was a fair criticism. Our own M3 analysis (Section 6.3) shows that the root cause is in the receiver, and that any separation between bursts hides it: one zero sample works as well as rolloff 4. Changing the rolloff in the example therefore hides a receiver bug instead of fixing it, and does

not help users whose bursts come from another transmitter. We agree, and propose the following revision of the same PR:

Proposed change	Reason
Drop the ofdm_loopback example commit	The example should not hide a receiver bug
Run the test with the default rolloff=0	The test must exercise GNU Radio's real default behaviour
Mark the test @unittest.expectedFailure, with a docstring referencing #6891	Documents the bug in the test suite without breaking CI. When the receiver is fixed, the test becomes an "unexpected success", which fails CI, so whoever fixes the bug must remove the decorator
Keep the helper _txrx_multipacket unchanged	Reused by a future receiver-side fix

The test is deterministic: the test class's setUp() already calls random.seed(0). With seed 0 and rolloff 0 the test lost packets on both GNU Radio builds we used (M2 result: 4850 != 5000 bytes). The revision will be pushed to the same PR #8271 once the maintainer confirms the direction; the alternative is to pair the test with a real receiver fix. No other pull request has been submitted.

4.4 Technical Significance

- **Closes a coverage gap.** All existing qa_ofdm_txrx tests send a single packet, while #6891 only occurs between packets.
- **Shown to detect the bug.** Negative controls on two GNU Radio builds fail exactly as expected (Section 6.6).
- **Low risk.** The proposed revision is a test-only change, with no C++ or API change. ● **Ready acceptance test for the real fix.** The trigger-logging method and the replay model give a measurable target for a receiver change in header_payload_demux.
- **Contribution level (course guidelines): Level C, testing.** The example change (Level A/B) is to be dropped, as proposed in our response to the review; the receiver-side fix (Level B/D) is the proposed next step.

5. Experimental Setup

Item	Value
OFDM parameters	FFT 64, CP 16, 48 data + 4 pilot subcarriers (52 occupied), BPSK header and payload, CRC32, scrambling on
Packet sizes	50 B by default; 20-200 B (E2, E8, E8b); 100-1500 B (E10)
Channels	Ideal; AWGN 6-30 dB; 4 static exponential power-delay profiles (1, 3, 8, 14 taps); CFO 0-3.5 subcarriers; time-varying indoor channel with a 0.5 Hz reflector (E9)
Baselines	Stock TX (rolloff 0) = GNU Radio default; alternatives: rolloff 1, 2, 4, one zero sample (gap 1), eight zero samples (gap 8)
Execution	Streaming and offline (M2); offline for sample-exact bookkeeping (M3)
Repetitions	5 seeds per point (3 for E8b)
Software	GNU Radio 3.10.12.0 / Python 3.11 / Ubuntu 22.04 (M2); 3.10.9.2 / Python 3.12 / Ubuntu 24.04 (M3); E8 re-run independently on a second machine (40/40 runs identical)

6. Results

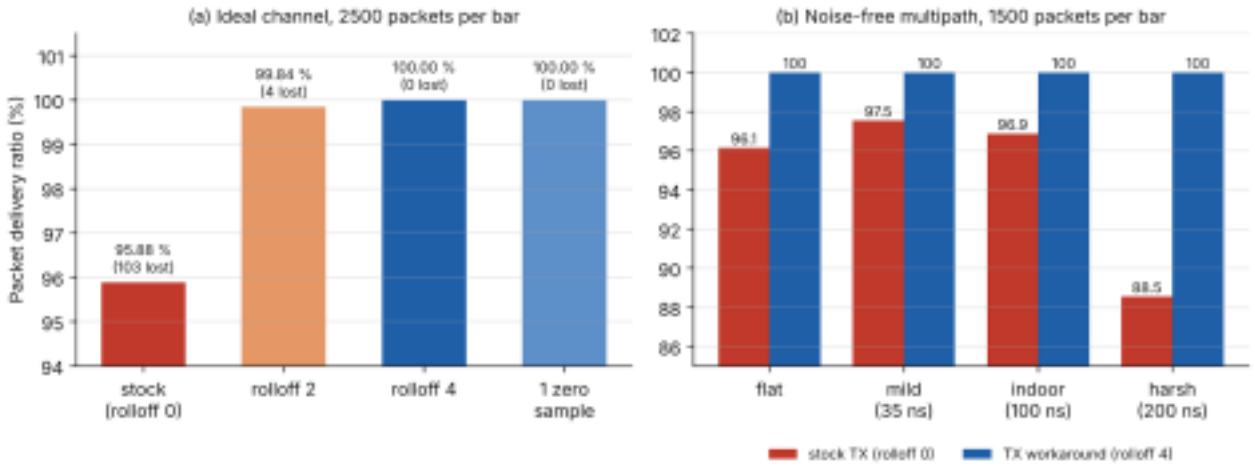


Figure 1. Summary of M2-M4 results. (a) Ideal channel, 50-byte packets, 5 seeds × 500 packets per bar. (b) Noise-free multipath, 5 seeds × 300 packets per bar.

6.1 The Loss and the Scheduler Question (E1)

TX setting	Samples between bursts PDR streaming	PDR offline Lost per seed (both modes)
rolloff 0 (default)	0 95.87 % 95.88 % 26, 21, 19, 18, 19	
rolloff 1	0 (GNU Radio resets it to 0) 95.88 % 95.88 % 26, 21, 19, 18, 19	
rolloff 2	1 non-zero tail sample 99.84 % 99.84 % 0, 2, 2, 0, 0	
rolloff 4	3 tapered samples 100 % 100 % 0 in every seed	
gap 1	1 zero sample 100 % 100 % 0 in every seed	
gap 8	8 zero samples 100 % 100 % 0 in every seed	

The lost-packet counts are identical in both modes for every seed, so the scheduler is not the cause. This answers a hypothesis raised in the #6891 thread.

6.2 Noise and Packet Length (E2, E3)

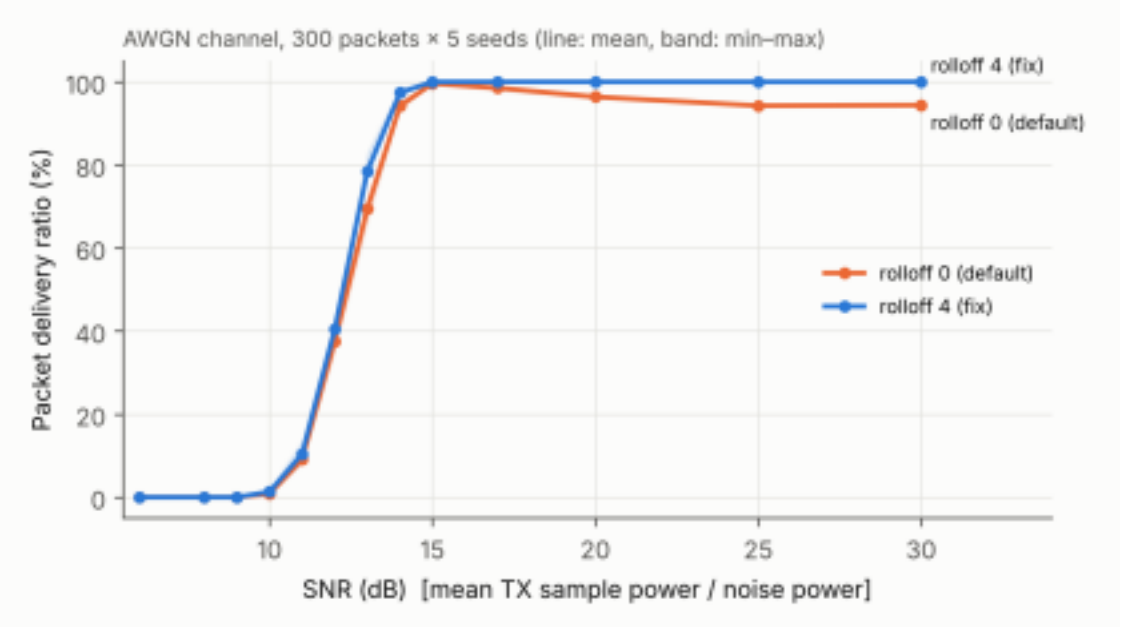


Figure 2. PDR vs SNR on an AWGN channel, 300 packets × 5 seeds per point.

SNR (dB)	10	12 14	15	17 20	25	30
Stock PDR (%)	0.9	37.5 94.2	99.6	98.5 96.4	94.3	94.4
rolloff 4 PDR (%)	1.3	40.5 97.4	100	100 100	100	100

Counter-intuitively, the stock loss **grows** with SNR: from 99.6 % delivery at 15 dB it falls to 94.4 % at 30 dB. Noise randomises the early-trigger error, while a clean channel makes it systematic. A test run with some noise can therefore pass while the bug is present. Packets of 20, 50 and 200 bytes lose about 4 %; 100-byte packets lose none (explained in Section 6.3).

6.3 The Mechanism at Sample Level (E8, E8b)



Figure 3. Range of Schmidl & Cox trigger positions relative to the true burst start (5 seeds × 500 packets per row). Red marks lost packets.

TX setting (50 B)	Lost (of 2500)	Trigger offset (samples)	Trigger spacing (samples)
rolloff 0 (stock)	103	69-73	956-964

rolloff 2	4	70-73	958-964
rolloff 4	**0**	**73 only**	**963 only**
gap 1	0	74 only	961 only

- **Finding 1.** With touching bursts, the random last symbol of the previous burst leaks into the correlation window, and the plateau detector fires anywhere from 69 to 73 samples after the true start. Any separation between bursts pins the trigger to a single position.
- **Finding 2.** The replay rule (Section 3.1) rejects **exactly** the lost packets, with the same sequence numbers, in 40/40 E8 runs and 60/60 E8b runs. Rolloff 2 adds only one sample of slack, so a trigger 3 or more samples early still loses the packet (4 of 2500).
- **Finding 3 (E8b, 20 packet lengths).** Loss occurs when the trigger jitter is 2 samples or more, and never when it is 1 sample or less (40, 45, 46, 70 and 100 B). The hypothesis "loss only when the last OFDM symbol is full" was **rejected**: 30 B and 60 B have partly filled last symbols and still lose 7 to 8 %. Consequence: a single-length test can pass by luck (for example at 100 B), which is why the upstream test uses 50 B.

6.4 Multipath, Noise and CFO (E6, E7)

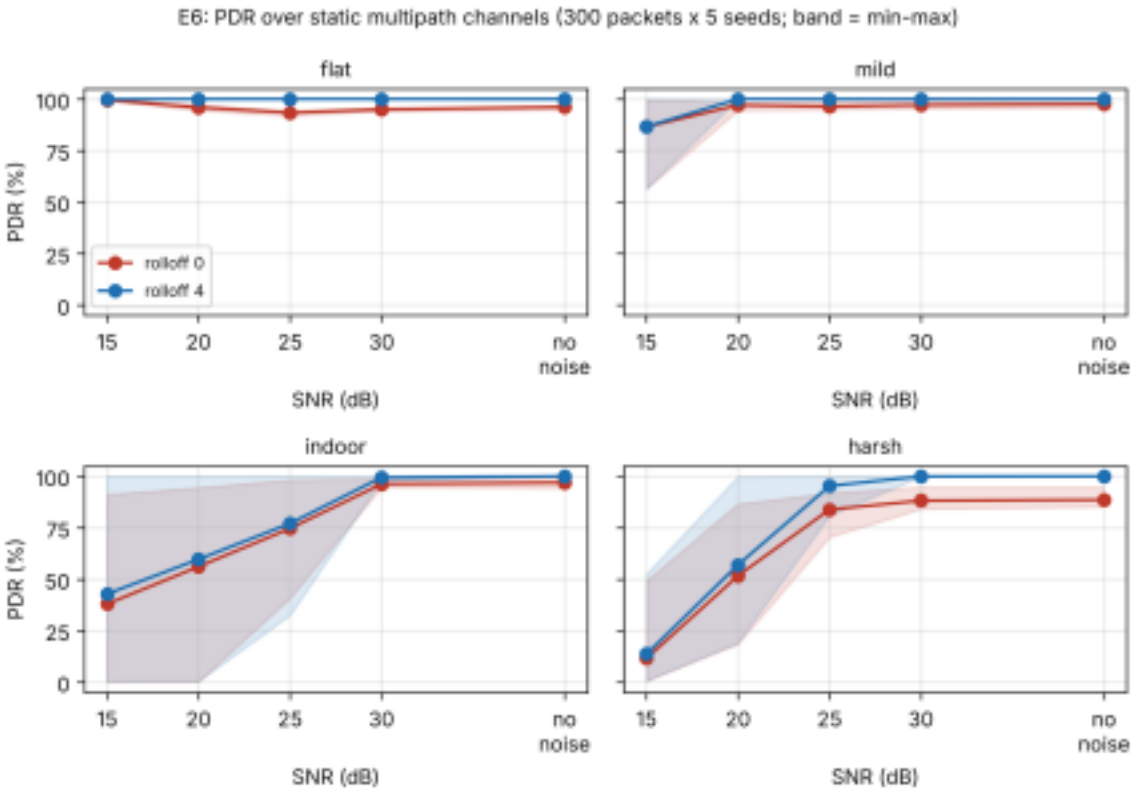


Figure 4. PDR vs SNR for four static multipath profiles. Red: stock (rolloff 0); blue: rolloff 4; shaded band: minimum to maximum over 5 seeds.

Profile (rms delay)	**20 dB: stock → rolloff 4 30 dB: stock → rolloff 4**	**Noise-free: stock → rolloff 4**
flat (0 ns)	95.9 → 100.0 94.9 → 100.0	96.1 → 100.0
mild (about 35 ns)	97.1 → 100.0 97.1 → 100.0	97.5 → 100.0
indoor (about 100 ns)	56.1 → 59.7 96.2 → 99.4	96.9 → 100.0

harsh (about 200 ns)	51.7 → 56.9 88.1 → 99.9	88.5 → 100.0

- The workaround is **never worse** in 20 of 20 profile and SNR combinations (+0.1 to +11.8 percentage points).
- **Multipath makes the bug worse.** Noise-free loss rises from 3.9 % (flat) to 11.5 % (harsh), because a longer impulse response smears the previous burst further into the next preamble. ● At low SNR on frequency-selective channels both settings collapse (indoor, 15 dB: 37.9 % vs 42.8 %). That is a limit of the uncoded BPSK + CRC32 design, not of this bug.
- **CFO** from 0 to 3.5 subcarrier spacings changes nothing (flat channel, 25 dB: stock 93.4 to 94.6 %, rolloff 4 100 % at every CFO).

6.5 What the Loss Means for Wi-Fi Sensing (E9)

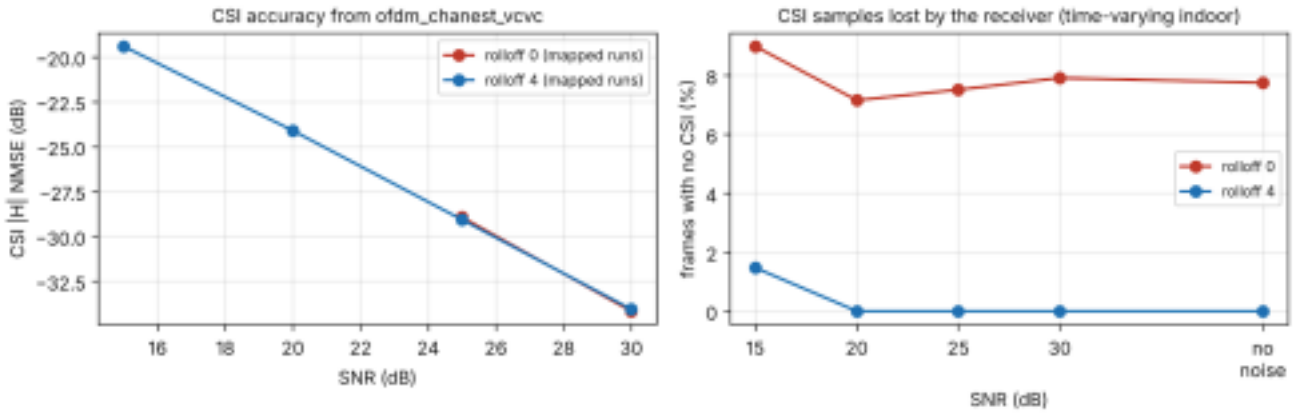


Figure 5. Left: accuracy of GNU Radio's CSI estimate (ofdm_chanest_vcvc) vs SNR. Right: fraction of frames without any CSI.

SNR	TX setting	Frames without CSI (of 503)	CSI NMSE	Motion-frequency error (timestamps from packet order)
25 dB	stock	37.8 (7.5 %)	−28.9 dB	**8.31 %**
25 dB	rolloff 4	0	−29.1 dB	0.06 %
30 dB	stock	39.8 (7.9 %)	−34.2 dB	**7.63 %**
30 dB	rolloff 4	0	−34.1 dB	0.06 %
noise-free	stock	39.0 (7.8 %)	< −60 dB	**7.40 %**
noise-free	rolloff 4	0	< −60 dB	0.00 %

- GNU Radio already computes accurate CSI (NMSE −19 dB at 15 dB SNR, −34 dB at 30 dB), but ofdm_rx does not output it: the channel-tap tag does not reach the block's output.
- With the stock TX, 7 to 9 % of CSI samples are silently missing on this channel. A sensing pipeline that assumes evenly spaced samples then over-estimates a 0.5 Hz motion by 7 to 8 %. ● With timestamps derived from sequence numbers, or with the TX workaround, the error is below 0.1 %.
- Caveat: with the stock TX at 25 dB and below, CSI frames could be matched to packets only in some runs (1 of 5 at 25 dB, 3 of 5 at 30 dB); cells without a matched run are not reported.

6.6 Freezes, Large Packets (E4, E5, E10) and Contribution Tests

Packet size	Default buffers	Hier-block buffers set to 65536

50-250 B	OK	OK
300 B	**HANG** (ofdm_cyclic_prefixer: "Buffer too small for min_noutput_items")	OK, 20/20
400-1000 B	**HANG** (ofdm_carrier_allocator / tagged_stream_mux)	OK, 20/20
1020 B and above	Finishes normally, **0 packets sent**	0 packets sent

- **Freeze mechanism.** In every HANG the failing block logs an error and its thread exits; the rest of the flowgraph waits forever and run() never returns. This is the symptom reported in #7997, reproduced on two GNU Radio builds. A transmitter length tag that leaks to the receiver causes the same freeze through header_payload_demux (E5).
- **New silent failure.** For packets of 1020 bytes or more, tagged_stream_mux logs ninput_items_required = 12032, max_possible_items_available = 8191, and the TX outputs nothing, without an exception, even with large buffers. Wi-Fi-MTU-size packets therefore cannot be sent through the stock chain.

Contribution tests.

Test	**GNU Radio build Result**
Submitted test (PR #8271), rolloff 4	3.10.12.0 (M2) 6/6 qa_ofdm_txrx tests OK in 3 repeated runs
Submitted test with rolloff 0 (negative control)	3.10.12.0 (M2) FAIL: 4850 != 5000 bytes (3 packets lost), so the test detects the bug
Submitted test + negative control + 20/50/100/200 B × 3 seeds	3.10.9.2 (M3) 3/3 OK
Proposed revised test (rolloff 0, expected failure)	Run by the PR author before pushing: Not run yet; expected output: OK (expected failures=1)

7. Baseline Comparison

Approach	**Where the change is**	**PDR, ideal channel (50 B)**	**PDR, harsh multipath, noise-free**	**Overhead**	**Limitation**
Stock GNU Radio (baseline)	None	95.88 %	88.5 %	0	Silent loss
rolloff 1	TX	95.88 %	n/a	0	GNU Radio resets it to 0
rolloff 2	TX	99.84 %	n/a	1 sample per 960	Still loses packets
rolloff 4 (submitted in PR #8271)	TX	**100 %**	**100 %**	3 samples per 960 (0.3 %)	Hides the RX bug; other transmitters still affected
gap 1 / gap 8 (zero samples)	TX	100 %	n/a	1 or 8 samples	Same as rolloff 4; needs an extra block

Regression test (proposed revision)	Tests only	Unchanged	Unchanged	0	Detects the bug, does not fix it
Receiver-side fix (proposed, GSoC 2027)	RX (header_payl oa d_demux or timing refinement)	Target 100 % with rolloff 0	Target 100 %	To be measured	Not yet implemented

All TX-side settings were compared on the same packets, seeds and channels. No receiver-side method has been implemented or measured, so we make **no claim** of improvement over a receiver fix.

8. Maintainer and Community Feedback, and Actions Taken

Date	Who	Feedback	Our action
Feb 2024	marcusmueller (maintainer)	Confirmed #6891 as "a rather serious" bug	Quantified it and wrote a test (M2)
Jun 2026	ffu (contributor)	Plateau + demux explanation; TX rolloff as the least invasive change	E8 confirmed the explanation to the sample; PR #8271 as submitted used the rolloff change
Jun 2026	Wiktorijus	Suspected buffers or the scheduler	E1 (streaming = offline) ruled this out
11 Nov 2026	Team G1	Measurements posted on #6891 and PR #8271 opened	-
12 Nov 2026	marcusmueller	On #8271: "are you proposing to abuse the rolloff factor to hide a bug?"; label "Needs Action"	We agree: the rolloff is a workaround and the root cause is in the receiver. We prepared a revision of the same PR (drop the example change; test with rolloff 0 and expectedFailure); the alternative is pairing the test with a receiver fix. Not pushed yet; waiting for the maintainer to confirm.

What we learned from the review. Our M3 report already listed "the fix is a workaround at the TX" as a limitation, yet we still submitted it as a fix. A maintainer's first concern is not "does it work" but "does it hide a problem that will affect someone else". Because we had trigger logs and a predictive model, we could answer the question with data instead of starting over.

9. Final SOTA Positioning

Strongest existing approaches. (i) Controllable OFDM PHY: GNU Radio gr-digital OFDM and gr-ieee802-11 [9]. (ii) CSI: chip-based tools (Intel 5300, Nexmon, ESP32, PicoScenes) [10, 11]. (iii) Sensing: SenseFi-style deep models and 802.11bf procedures [6, 8]. (iv) Timing: Schmidl & Cox [3] and its refinements.

Work	Their state	After Team G1's work

GNU Radio 3.10 gr-digital OFDM	Bug known since 2023, cause described qualitatively, no multi-packet test	Root cause measured to the sample, with a model that predicts every lost packet; scheduler ruled out; first multi-packet regression test under review upstream; three silent failure modes documented with reproducers
GNU Radio 4.0	New runtime and buffers	The drop is caused by signal processing, so it will also exist in GR4 unless the blocks change; our tests are runtime-agnostic checks
gr-ieee802-11	Standard PHY with its own synchronisation	Our trigger-logging and replay method can measure its packet-level reliability directly
Chip-based CSI,	Real hardware, closed PHY	GNU Radio already computes accurate CSI (−34 dB NMSE at 30 dB) but does not expose it; an open, controlled CSI source is feasible.
Tools SenseFi, 802.11bf	Assume uniformly sampled, timestamped CSI	We quantify how a 7-9 % silent drop violates that assumption (7-8 % frequency bias) and argue for CSI output with packet numbers

Key technical contributions.

1. A measurement method: sequence numbers, two execution modes, watchdog, trigger logging and a replay model.
2. A sample-exact, predictive explanation of #6891.
3. An evaluation of the TX workaround over multipath, SNR, CFO and two GNU Radio builds. 4. The first quantitative link between GNU Radio's OFDM receiver reliability and CSI-sensing error. 5. One upstream pull request (#8271) adding a regression test, with a revision prepared after the maintainer's review.
6. A new silent failure mode (packets of 1020 bytes or more) and the freeze mechanism behind the #7997 symptom.

What we do not claim. No new algorithm, no receiver fix implemented, no merged code yet, no hardware validation.

Remaining gaps and what would close them.

Gap	**What is needed**
Receiver still drops packets when bursts touch	Early-trigger tolerance in header_payload_demux, or fine timing refinement using the known second sync word; accept only with 100 % PDR on our test matrix and no QA regression
Contribution not yet merged	Agree the revision with the maintainer and push it to PR #8271; build GNU Radio main from source and run the OFDM QA tests
CSI not exposed by ofdm_rx	Optional per-packet CSI message port with packet number and timestamp, plus an example that writes a dataset
Silent freezes and the 1020-byte refusal	Clear errors or graceful drops instead of thread exit; buffer sizing fix or a documented limit; a new issue with our reproducer
Simulation only	Repeat E6 and E9 on SDR hardware (two PlutoSDRs or USRPs) or in GNU Radio's hardware-in-the-loop CI

10. Record of Open-Source Contributions

Item	Link / identifier	Status
Issue comment with measurements on #6891	https://github.com/gnuradio/gnuradio/issues/6891#issuecomment-6787055390	Posted 11 Nov 2026
Pull request	https://github.com/gnuradio/gnuradio/pull/8271	Open; label "Needs Action"; this is our only pull request
Revised test (proposed)	Prepared by the team	Not pushed yet; goes into PR #8271 after the maintainer confirms
Comment on #7997	-	Not posted; freeze mechanism documented in this report (Section 6.6)
New issue for the 1020-byte silent refusal	-	Not filed (future work)
Team repository	https://github.com/divyarajv1-ship-it/g1_gsoc_wifisensing	M1-M4 folders, commits by all five members

Code quality. Signed-off commits (DCO passed), GNU Radio commit-message style (module: component: summary, Refs #6891), test written in the existing gr_unittest style reusing the file's helpers, no C++ or API change.

11. Limitations and Challenges

1. **The contribution is a test, not a fix.** The receiver still drops packets when bursts touch. 2. **Simulation only.** Synthetic channels (exponential power-delay profiles, a sinusoidal reflector); real SDRs add sample-rate offset, phase noise and AGC effects.
3. **#7997 not confirmed on the reporter's own flowgraph**, which is not public; we show a mechanism with the same symptom on two builds.
4. **E8b only partly explained.** Jitter predicts loss, but why certain packet lengths produce no jitter is still open.
5. **Low-SNR frequency-selective performance** is limited by uncoded BPSK; this is out of scope. 6. **CSI matching with the stock TX at low SNR** was not always possible (E9 caveat). 7. **Not tested on a source build of GNU Radio main**, only on the packaged releases 3.10.9.2 and 3.10.12.0.
8. **Challenges.** Freezes that never return (solved with a process watchdog); losses hidden by repeated payloads (solved with sequence numbers); learning the upstream process (DCO, PR template, review etiquette).

12. Conclusion and Future Work

We turned a two-year-old "serious" GNU Radio bug from a qualitative description into a measured, predicted and tested failure, linked it to a measurable error in Wi-Fi sensing, and took it upstream. The maintainer's review led us to prepare a revision that turns our contribution from a workaround into a test that documents the bug, which reflects the real state of the code today. The next step, which needs C++ changes and maintainer agreement, is a receiver-side fix plus a supported CSI output. This is the basis of our GSoC 2027 proposal (Appendix A).

Future work.

1. Push the prepared revision to PR #8271 once the maintainer agrees, and work towards getting it merged.
2. Prototype and compare the two receiver-side options on the M3 test matrix.

3. Report the 1020-byte refusal as a new issue and propose "fail loudly" checks.
4. Add a per-packet CSI output with packet numbers and timestamps.
5. Validate on SDR hardware.

13. Reflection on the Open-Source Contribution

The project started as "fix a freeze" (#7997) and ended as "make a silent packet loss measurable and tested" (#6891), because #7997 could not be reproduced exactly without the reporter's flowgraph, while #6891 could. The most useful thing we contributed is not the one-line rolloff change but the evidence: a model that predicts every lost packet, and a test that fails on the bug. The review taught us that in a mature project a small change that hides a problem is worse than no change, and that a contributor's job is to make the maintainer's decision easy with data. Today's impact on the ecosystem is a documented root cause and a test under review; the impact we aim for in GSoC 2027 is a receiver that does not lose packets and that exposes its CSI.

14. Individual Contributions (Milestone 4)

Member	Milestone 4 work	Evidence
Hirva Vekariya	Managed the upstream work for M4: prepared the response to the maintainer's review on PR #8271 and the revised test (rolloff 0 with expectedFailure); integrated the final report and the refined GSoC 2027 proposal	https://github.com/divyarajv1-ship-it/g1_gsoc_wifisensing/commit/b3a91ed71c926cb7067eb2b9c550e2b2a4a8c9dd
Tirth Pathar	Prepared the final experiment code for M4: cleaned all experiment scripts (E1-E10) so they run from one script, and added the environment file and the 2-minute smoke test	https://github.com/divyarajv1-ship-it/g1_gsoc_wifisensing/commit/70f78b2
Divyaraj Vaghela	Set up the M4 submission folder and its README, organised the repository for the final submission, and updated the top-level README with the M4 summary and final project status	https://github.com/divyarajv1-ship-it/g1_gsoc_wifisensing/tree/main/M4_g1_gsoc_wifisensing
Jahnavi Patel	Consolidated the M2-M4 results and figures in one place, and added the M4 summary analysis that produces Figure 1 and the consolidated results tables	https://github.com/divyarajv1-ship-it/g1_gsoc_wifisensing/commit/3a39ff1
Jinay Shah	Wrote the data documentation, prepared the video slides and script, recorded the 20-minute video walkthrough and published the video link	https://github.com/divyarajv1-ship-it/g1_gsoc_wifisensing/commit/6bbd52e886c02fdbe6884eebf631c9214771de57 https://github.com/divyarajv1-ship-it/g1_gsoc_wifisensing/commit/e29d50ea6bb22293bf98840b4164ebe55201ea9c

15. Reproducibility

The team repository contains one script that re-runs all experiments, both contribution tests and all figures in about 35 minutes, and a 2-minute smoke test (stock TX, seed 1 must lose 26 of 500 packets; rolloff 4

must lose 0). The environment can be created with conda/micromamba without administrator rights, or with the Ubuntu 24.04 GNU Radio package. All data are generated by the scripts with fixed seeds; no external dataset is needed.

References

1. GNU Radio Issue #7997, "Complete OFDM Flowgraph Freeze After Inserting Channel Block in GNU Radio," 2026. https://github.com/gnuradio/gnuradio/issues/7997
2. GNU Radio Issue #6891, "OFDM Loopback Example drops packets even with 0 noise in simulation," 2023. https://github.com/gnuradio/gnuradio/issues/6891
3. T. M. Schmidl and D. C. Cox, "Robust frequency and timing synchronization for OFDM," IEEE Trans. Commun., vol. 45, no. 12, pp. 1613-1621, 1997.
4. GNU Radio source code: gr-digital header_payload_demux, ofdm_txrx and qa_ofdm_txrx; gnuradio-runtime block_executor. https://github.com/gnuradio/gnuradio
5. Y. Ma, G. Zhou and S. Wang, "WiFi sensing with channel state information: A survey," ACM Computing Surveys, vol. 52, no. 3, 2019.
6. R. Du et al., "An overview on IEEE 802.11bf: WLAN sensing," IEEE Communications Surveys & Tutorials, vol. 27, no. 1, 2025.
7. A. Y. Radwan et al., "A Tutorial-Cum-Survey on Self-Supervised Learning for Wi-Fi Sensing," IEEE Communications Surveys & Tutorials, vol. 28, 2026.
8. J. Yang et al., "SenseFi: A Library and Benchmark on Deep-Learning-Empowered WiFi Human Sensing," Patterns, vol. 4, no. 3, 2023.
9. B. Bloessl et al., gr-ieee802-11. https://github.com/bastibl/gr-ieee802-11
10. D. Halperin et al., "Tool release: Gathering 802.11n traces with channel state information," ACM SIGCOMM CCR, vol. 41, no. 1, 2011.
11. F. Gringoli et al., "Free your CSI: A channel state information extraction platform for modern Wi-Fi chipsets," ACM WiNTECH, 2019.
12. V. Erceg et al., "TGn channel models," IEEE 802.11-03/940r4, 2004.
13. GNU Radio pull request #8271. https://github.com/gnuradio/gnuradio/pull/8271
14. Team G1 comment with measurements on GNU Radio issue #6891, 11 Nov 2026. https://github.com/gnuradio/gnuradio/issues/6891#issuecomment-6787055390

Appendix A. Refined GSoC 2027 Proposal (Summary)

Title. Reliable, CSI-producing OFDM receiver for GNU Radio. **Size: 350 hours, medium difficulty. Everything in the table below is planned work for GSoC 2027; none of it has been done or submitted yet.**

What changed since the M3 draft. The maintainer's review of #8271 made clear that a transmitter-side workaround will not be accepted as the fix. The proposal therefore puts the **receiver-side fix first**, keeps the test-first approach, and moves the CSI output after it. The TX-rolloff fallback from the M3 draft has been removed.

Phase	Weeks	Deliverable
Community bonding	-	Design note agreed with mentors; source build of main; finish the review of our existing PR #8271
Tests and benchmark	1-2	Multi-length, multi-channel PDR test matrix as QA and benchmark scripts
Receiver fix	3-6	Prototype early-trigger tolerance and fine timing refinement; compare PDR and CPU cost; implement the chosen fix; remove expectedFailure (midterm goal)
Loud failures	7-8	Maximum-payload check in header_payload_demux; clear errors instead of thread exit; fix or documented limit for packets of 1020 bytes or more

CSI output	9-10	Optional per-packet CSI message port (packet number, timestamp, CSI vector) and a dataset example
Validation and documentation	11-12	CSI vs ground truth; tutorial page; final report

Acceptance criteria for the receiver fix. With rolloff 0 at the TX: PDR = 100 % on the test matrix (packet lengths 20-1000 B; flat, mild, indoor and harsh multipath; CFO up to 3.5 subcarriers; SNR 25 dB and above); the expectedFailure decorator on the #8271 test can be removed; no regression in existing QA tests; CPU cost measured and reported.